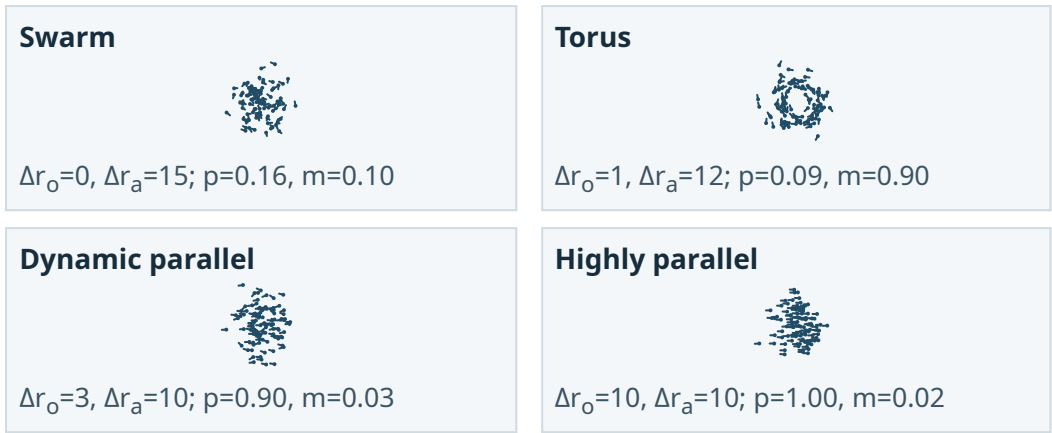


How a school swarms

Why fish swarm, circle and swim together

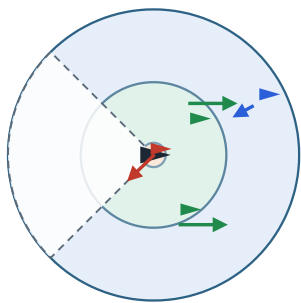
✓ Claim checked in source · 📍 Our model, code or measurement
BL = body length. p = heading alignment; m = group rotation. Both range from 0 to 1.

A Four collective states



📍 100 simulated fish. These zone settings are our examples.

B Three zones, one blind spot



- Repulsion**
Turn away; highest priority.
- Orientation**
Match neighbouring headings.
- Attraction**
Move towards neighbours.
- Blind spot:** 90° behind.

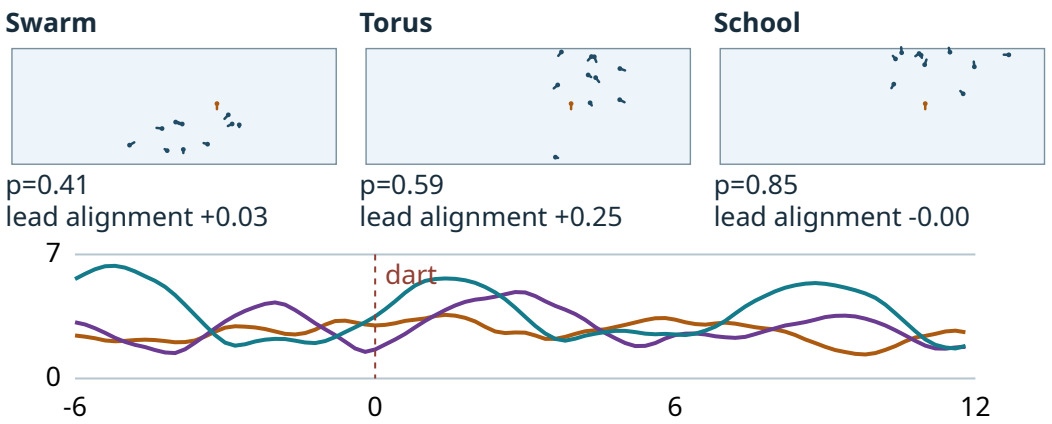
✓ S1: $r_f=1$ BL, $\alpha=270^\circ$. 📍 Shown: $\Delta r_o=5, \Delta r_a=6$.

D From neighbours to movement

- ✓ See neighbours within α . sch-pair
- ✓ Inside r_f ? Repel first. sch-pair
- ✓ Else combine heading and attraction. sch-want
- ✓ No neighbour? Keep the heading. sch-want
- ✓ Turn $\leq \theta_t$; move st. sch-turn
- 📍 Give the leader extra weight. weight
- 📍 Repel at walls; clamp position. sch-walls
- 📍 A dart startles nearby fish. sch-startle-all

✓ S1: $s=3$ BL/s, $\theta=40^\circ/s$. 📍 Game: 2 BL/s, 120°/s; no model noise. A 90° turn travels 6.75 vs 1.50 BL.

G The model inside the tank



Seconds from dart; distance in BL. **Swarm** · **torus** · **school**.

📍 Tank $13.4 \times 3.4 \times 4.7$ BL; $w=3$. Means: 3 seeds; drawing: 1. Level p: 0.39→0.77; lead alignment +0.40. Dart: 1.55→4.17 BL in 1 s, 2.4 BL after 4 s.

3,324 B

5.1% of 64 KB; 210 lines.

7.0 → 11.3 ms

Bare → engine tick, 11 fish; earlier 2,936 B version.

7.3 ms

Bare interpreter; error 1e-03 vs Python.

59.9 fps

HD release; p90 33.4 ms. Director 6.4 ms profiled.

📍 Mailbox calls: 4.2→0.28 μ s after debug-output fix. Historical timings.

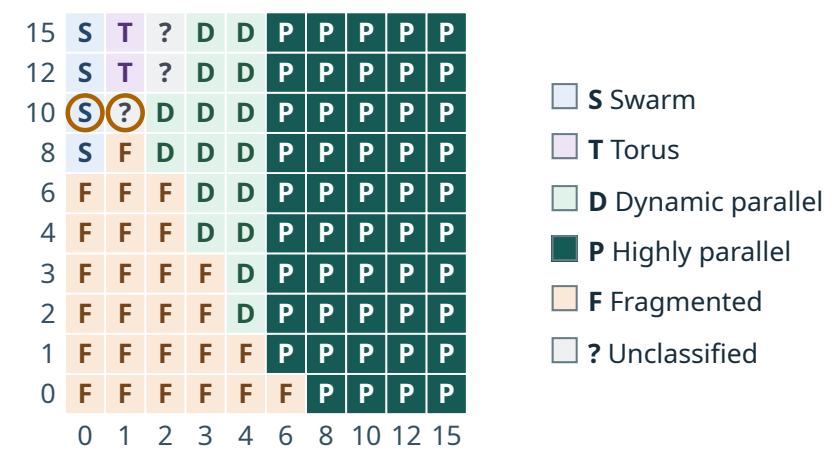
Sources and scope

S1 ✓ [Couzin et al. 2002, J. Theor. Biol.](#)
zone model, states, metrics, Fig. 3

S3 ✓ [Pitcher 1998, Shoaling and Schooling](#)
schooling: coordinated swimming within a shoal

📍 S5: aquarium code and recorded runs. Schooling is a game choice; real ocellaris live in anemone-associated groups. The model's lead remains untuned. Data 2026-10-01; sources checked 2026-10-06.

C A map of collective states



Across: Δr_o (BL). Up: Δr_a (BL).

📍 N=100; 3 replicates $\times$ 700 steps. Outlines: tank settings. Hysteresis untested.

F The Forth implementation

📍 Four complete school.fth words; comments omitted.

```
: sch-pair ( j -- )
  dup MB_YOU sc!
  sch-stride * sch-base +
  dup read-mailbox MB_PERP sc@ - MB_U sc!
  dup 1 + read-mailbox MB_PERP 1 + sc@ - MB_U 1 + sc!
  2 + read-mailbox MB_PERP 2 + sc@ - MB_U 2 + sc!
  MB_U vlen2 dup 0.000001 >
  over MB_R2 sc@ < &
  if
    sch-sqrt dup MB_DIST sc!
    1 swap / MB_U vscale
    heading-dot-u MB_COSB par@ >=
    if
      r-rep inside if
        -1 MB_SUM_R acc+u
        MB_N_R count+
      else r-ori inside if
        you-weight MB_SUM_O acc+v
        MB_N_O count+
      else
        you-weight MB_SUM_A acc+u
        MB_N_A count+
      then then
    then
  else drop then ;
```

```
: sch-want
  MB_WANT vzero
  MB_N_R seen? if
    MB_SUM_R add-wanted
  else
    MB_N_O seen? if MB_SUM_O add-wanted then
    MB_N_A seen? if MB_SUM_A add-wanted then
  then
  MB_WANT vnorm 0.000001 > not if
    MB_WANT heading>sc
  then ;
```

```
: sch-turn
  MB_VX me sch@ MB_WANT sc@ *
  MB_VY me sch@ MB_WANT 1 + sc@ * +
  MB_VZ me sch@ MB_WANT 2 + sc@ * +
  dup MB_COST par@ >= if drop else
    perp-all
    MB_PERP vnorm 0.000001 > not if
      perp-fallback
    then
    turn-all
    MB_WANT vnorm drop
  then ;
```

```
: sch-follow
  dup MB_ME sc!
  clear-sums
  sch-scan-cache
  sch-n 0 do
    i over <> if i sch-pair then
      loop
      drop
      sch-walls sch-want
      sch-startle? if
        sch-startle-away
        sch-startle-tick
      then
      sch-turn
      MB_WANT sc@ MB_NVX me sch!
      MB_WANT 1 + sc@ MB_NVY me sch!
      MB_WANT 2 + sc@ MB_NVZ me sch!
      0 advance 1 advance 2 advance
      clamp-all ;
```

S2 ✓ [Couzin et al. 2005, Nature](#)
simple model: informed minority guides a group

S4 ✓ [Madduppa et al. 2014, PLOS ONE](#)
adults in anemone-associated social groups